SUBSCRIPTION MANAGEMENT · SALESFORCE TO AZURE

High-level design

Agent 02 · Modelling · arc42 with C4 views

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Contents

1. Introduction and goals
2. Constraints
3. Context and scope (C4 level 1)
4. Solution strategy
5. Building block view
    5.1 Domain map
    5.2 Containers (C4 level 2)
    5.3 Inside a domain
6. Runtime view: business processes
    6.1 Subscription lifecycle
    6.2 Won deal becomes a subscription
    6.3 Saving a subscription
    6.4 Subscription At Risk Alert
    6.5 Renewal reminders
    6.6 Case routing and escalation
    6.7 Working alongside Salesforce
7. Deployment view
    7.1 Target deployment on Azure
    7.2 Services and sizing

7.3 Environments and release
7.4 Availability and recovery
8. Cross-cutting concepts
8.1 Data model of the first domain
8.2 Security across domains
8.3 Migration waves and system of record
8.4 Integration and events
8.5 Adding the next module
8.6 Salesforce products outside the core org
8.7 Observability
8.8 Traceability
9. Architecture decisions
10. Quality requirements
11. Risks and technical debt
12. Glossary

Figures

Figure 1. Context and scope (C4 level 1)
Figure 2. Domain map
Figure 3. Containers (C4 level 2)
Figure 4. Inside a domain
Figure 5. Subscription lifecycle
Figure 6. Won deal becomes a subscription
Figure 7. Saving a subscription
Figure 8. Subscription At Risk Alert
Figure 9. Renewal reminders
Figure 10. Case routing and escalation
Figure 11. Working alongside Salesforce
Figure 12. Target deployment on Azure
Figure 13. Data model of the first domain
Figure 14. Security across domains
Figure 15. Roles in the first domain
Figure 16. Load order in the first domain

Structure follows arc42; context and container views follow C4; decisions are recorded as ADRs. The platform is designed to receive every module of the Salesforce org; Subscription Management is its first domain.

1. Introduction and goals

Source	Salesforce org, 505 objects described
Platform scope	Every module of the org, domain by domain: Core customer, Sales, Service, Subscriptions, Activities
First domain	Subscription, Subscription Line (pilot)
Reference objects	Account, Contact, Opportunity, Product
Target	Microsoft Azure: Container Apps, Azure Database for PostgreSQL, Service Bus, Event Hubs, API Management, Front Door, Entra ID, Key Vault, Azure Monitor
Data in the first domain	Account 22 · Contact 20 · Opportunity 31 · Product 17 · Subscription 107 · Subscription Line 219

Quality goals, in priority order: **behavioural parity** with Salesforce, **security** at least equal to the org's, **modularity** so any module can be added or moved on its own, **availability** through a zone failure, **operability** with everything as code.

2. Constraints

- Each object has exactly one master at any time: Salesforce until its domain cuts over, Azure after.
- After the last wave the platform has no dependency on Salesforce.
- Salesforce record IDs are kept, so integrations, links and reports keep working.
- Platform services only (no virtual machines), in the client's own tenant and subscription; data stays in the chosen region.
- Open standards throughout (PostgreSQL, containers, OpenAPI, CloudEvents, OpenID Connect), so the platform is portable.

3. Context and scope (C4 level 1)

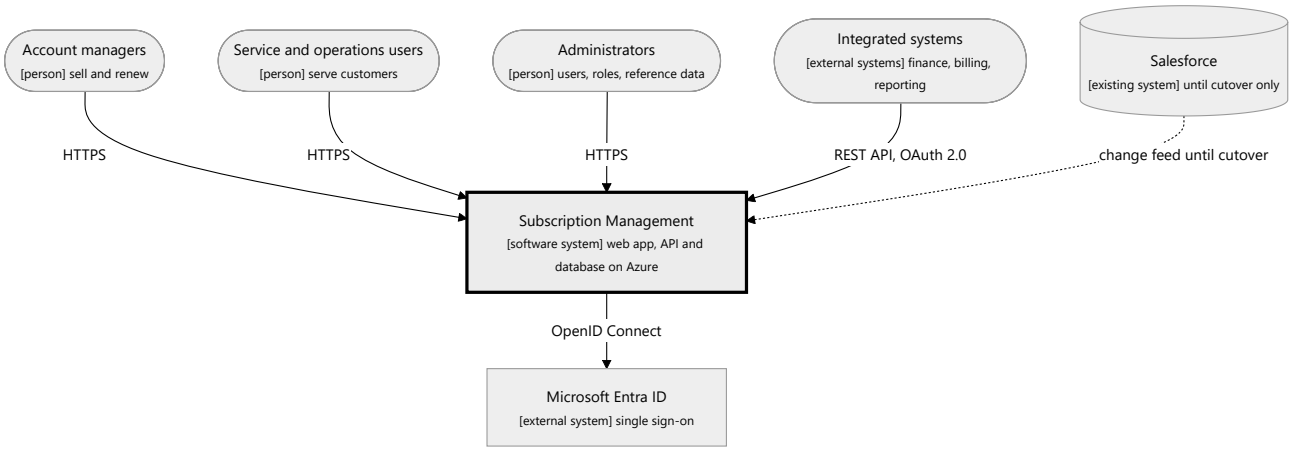


Figure 1. *Context and scope (C4 level 1)*

Users and integrated systems use the platform directly. Salesforce feeds changes to it only until each domain is cut over.

4. Solution strategy

Concern	Approach	Decision
Modules	Business domains around a shared core customer domain	ADR-015
Structure	Modular monolith: one deployable, one database, one schema per domain	ADR-016
Dependencies between domains	Published read views and events with a transactional outbox	ADR-017
Coexistence with Salesforce	One master per object; change feed and gateway routing	ADR-018
Security	One access model for every domain, enforced by row-level security	ADR-019, ADR-009
Identity	Entra ID for staff, External ID for portal users, managed identities for services	ADR-013
Hosting and network	Container Apps and PostgreSQL; private endpoints; Front Door with WAF	ADR-012, ADR-014
Portability	Open standards; only infrastructure templates are Azure-specific	ADR-020
Records, calculations and rules	Salesforce IDs as keys; calculations and rules in the database	ADR-002 to ADR-008

5. Building block view

5.1 Domain map

Domains and the links between them, counted from the relationships in this org. Every domain may use the core customer domain; other links go through read views and events.

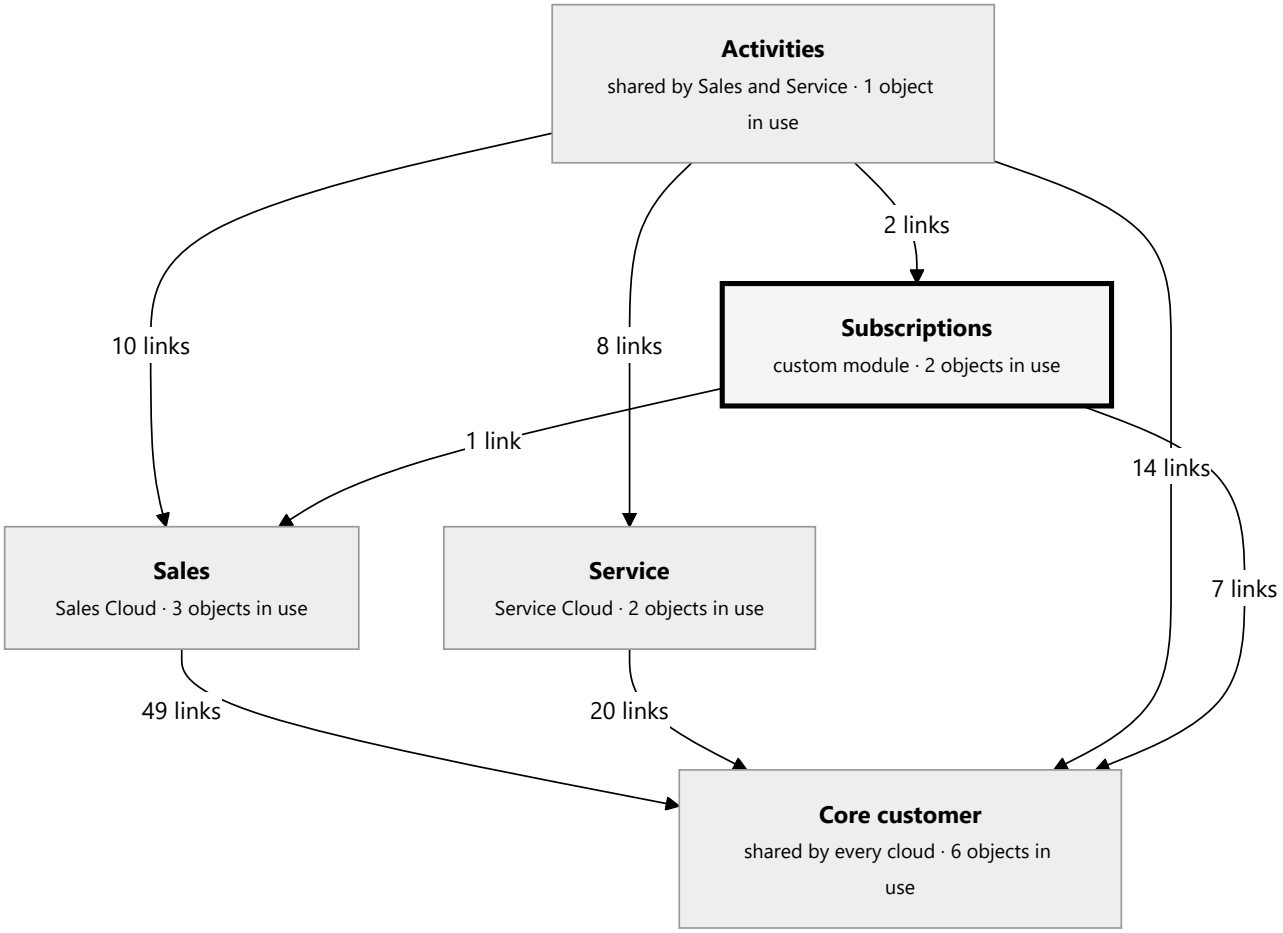


Figure 2. *Domain map*

Domain	Salesforce cloud	Objects in this org	Records	Depends on
Core customer	shared by every cloud	Account, Contact, User, Product, Price Book, Price Book Entry	103	—
Sales	Sales Cloud	Lead, Opportunity, Campaign	57	Core customer
Service	Service Cloud	Case, Entitlement	27	Core customer
Subscriptions	custom module	Subscription, Subscription Line	326	Core customer, Sales
Activities	shared by Sales and Service	Task	49	Core customer, Sales, Service, Subscriptions
Field service	Field Service	—	0	not in use in this org

5.2 Containers (C4 level 2)

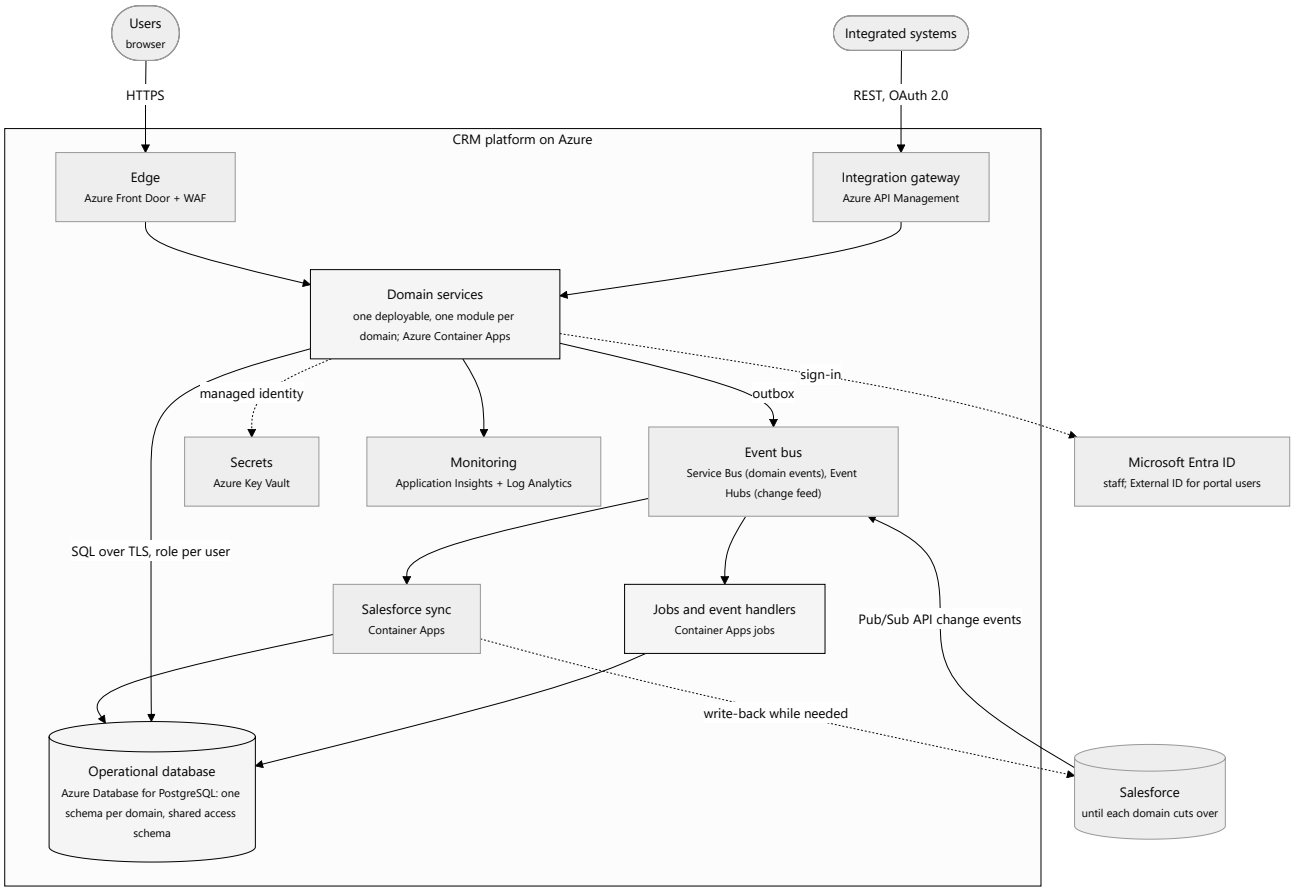


Figure 3. *Containers (C4 level 2)*

Container	Technology	Responsibility
Edge	Azure Front Door Premium with WAF	Single public entry point, TLS, protection against common web attacks
Integration gateway	Azure API Management	Salesforce-compatible REST API for integrated systems; routes each object to its current master; OAuth 2.0, throttling, versioning
Domain services	Node.js on Azure Container Apps	Screens and APIs of every domain, one module per domain; sets the database role of the signed-in user on every request
Jobs and event handlers	Container Apps jobs	Scheduled work such as renewal reminders; consumers of domain events
Salesforce sync	Container Apps	Applies the Salesforce change feed to read copies; writes back while Salesforce features still need it
Event bus	Service Bus Premium, Event Hubs	Domain events between domains; the Salesforce change feed
Operational database	Azure Database for PostgreSQL, Flexible Server	One schema per domain plus the shared access schema; rules, calculations and access enforced for every client
Secrets and monitoring	Key Vault; Application Insights and Log Analytics	Credentials read with managed identities; traces, metrics, logs, alerts and audit

5.3 Inside a domain

Every domain is built from the same template, so adding a module repeats a known pattern.

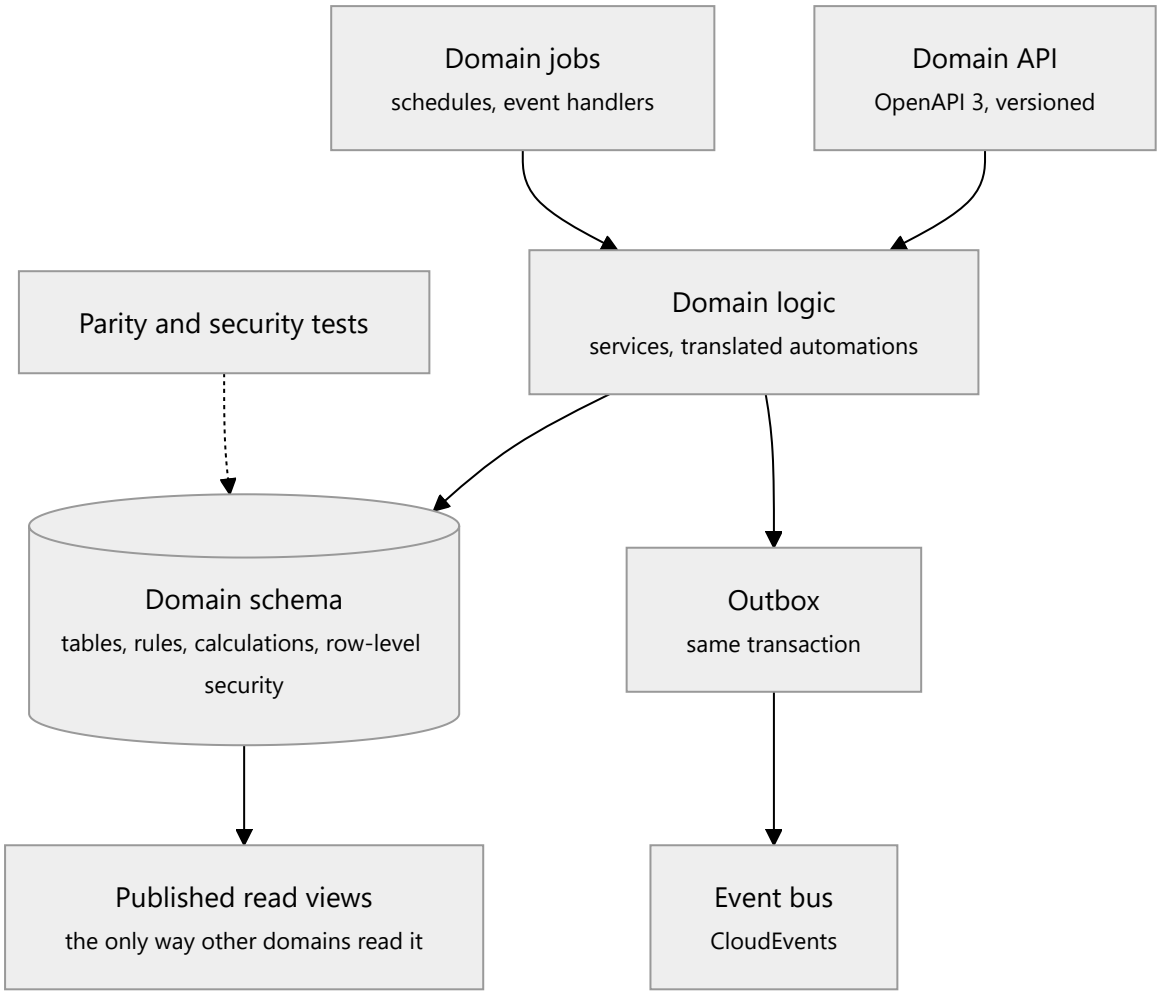


Figure 4. *Inside a domain*

6. Runtime view: business processes

6.1 Subscription lifecycle

Statuses come from the Status picklist; the notes come from the validation rules that guard each status.

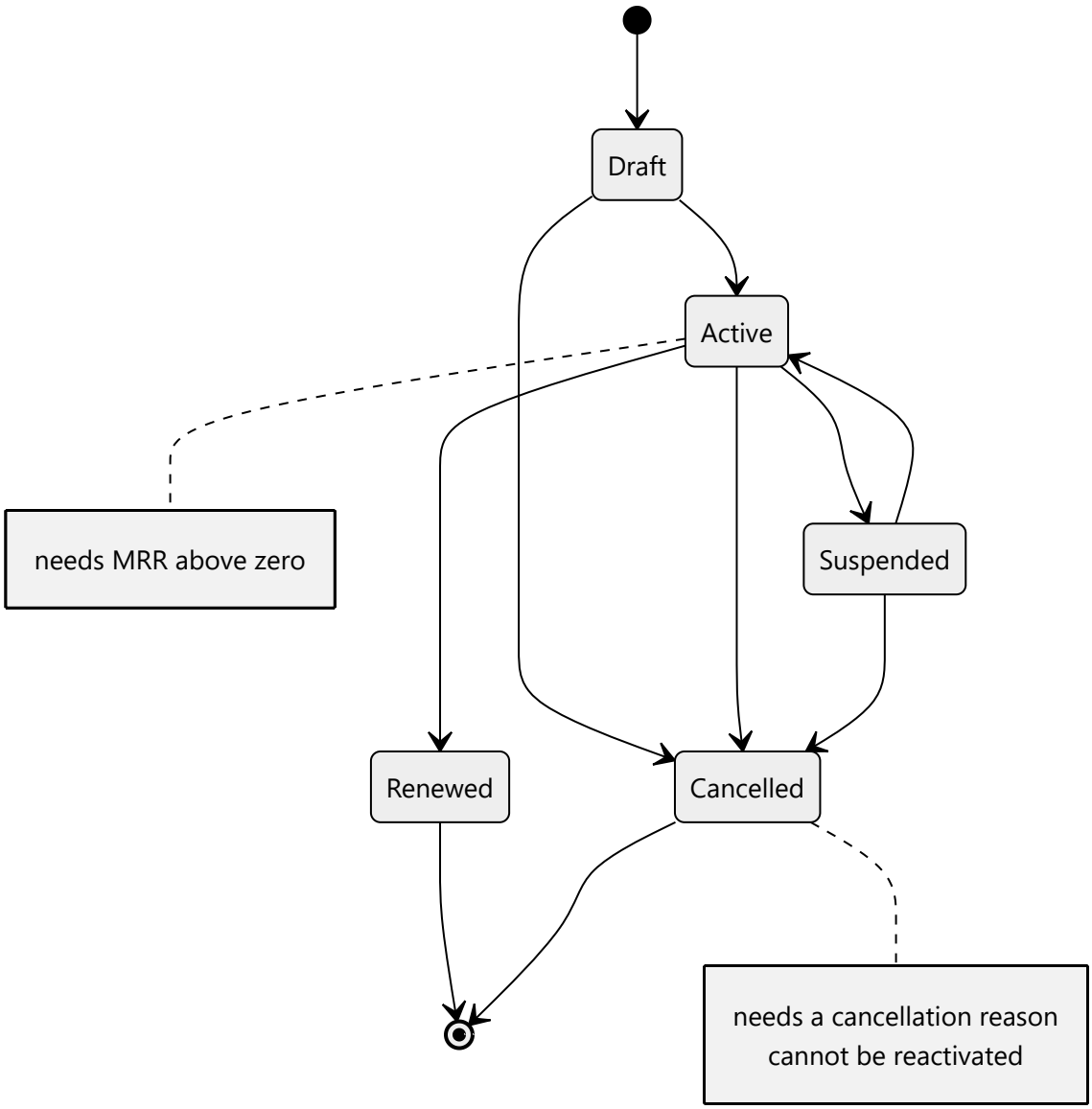


Figure 5. *Subscription lifecycle*

6.2 Won deal becomes a subscription

When an opportunity is marked Closed Won, a subscription is created for the account, once per deal. In the target, Sales publishes an "opportunity won" event and Subscriptions creates the subscription (section 6.7).

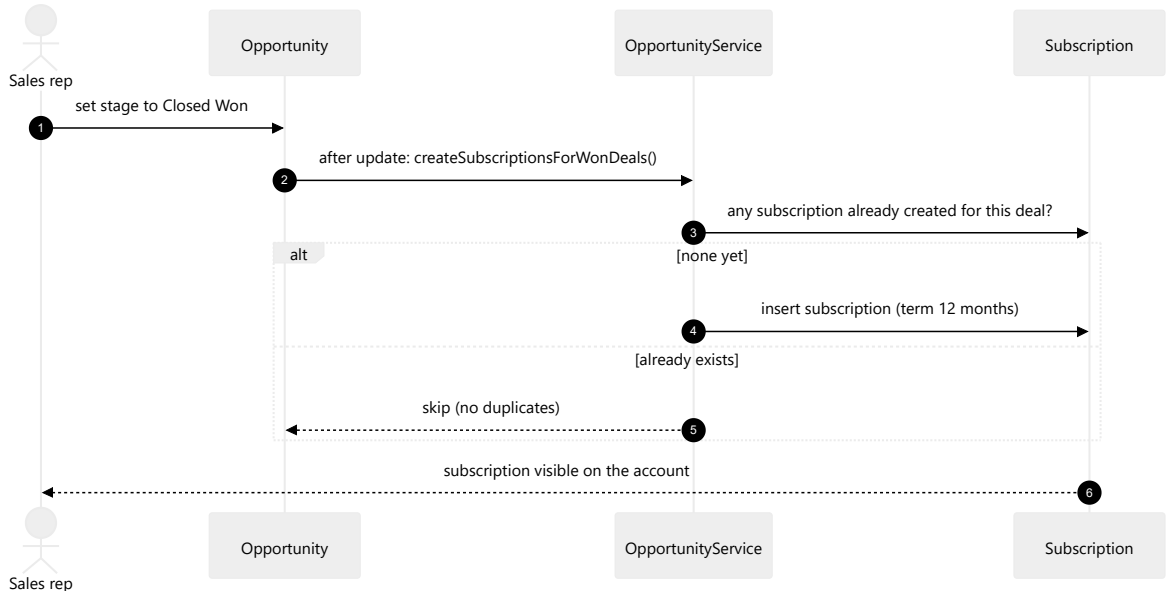


Figure 6. *Won deal becomes a subscription*

6.3 Saving a subscription

What happens, in Salesforce's order, every time a subscription is saved.

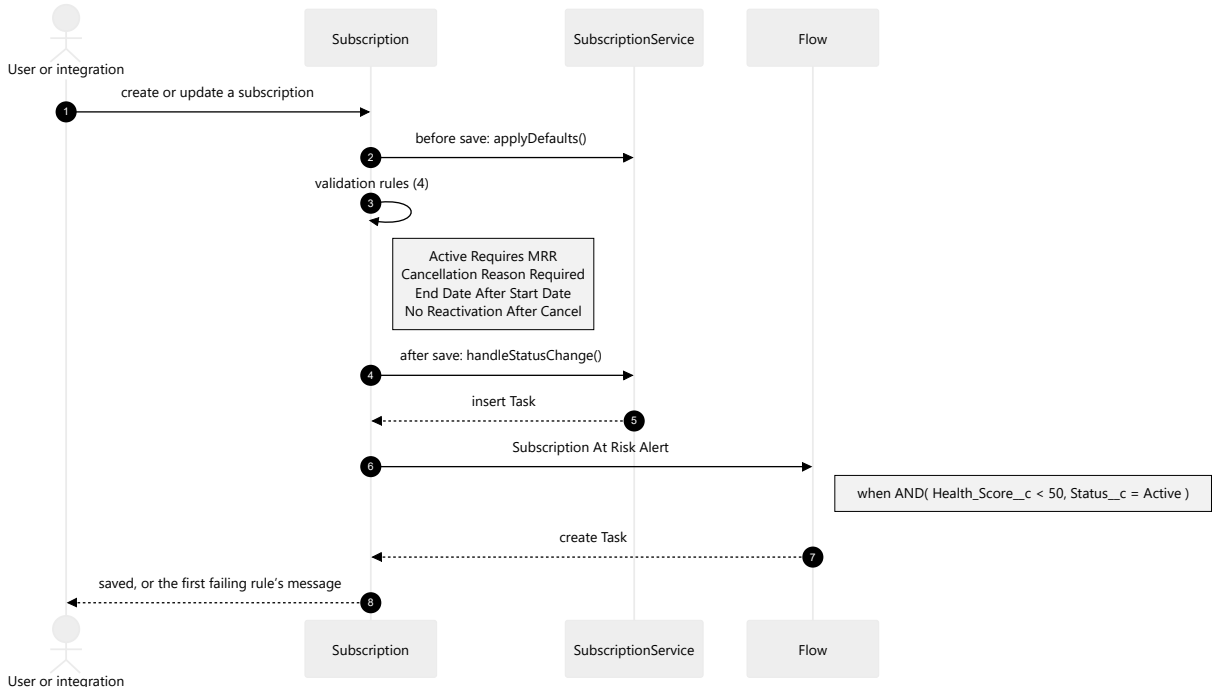


Figure 7. *Saving a subscription*

6.4 Subscription At Risk Alert

Raises a check-in task when an active subscription drops below the at-risk health threshold.

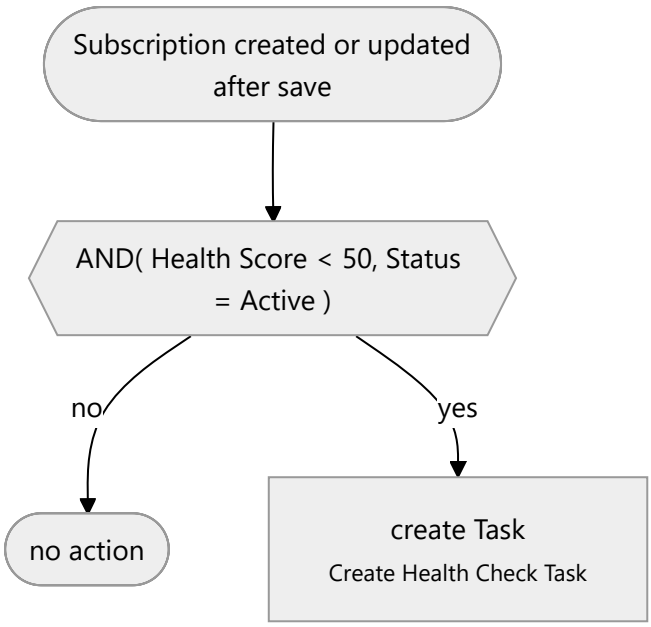


Figure 8. *Subscription At Risk Alert*

6.5 Renewal reminders

A scheduled job raises a renewal task before each active subscription ends.

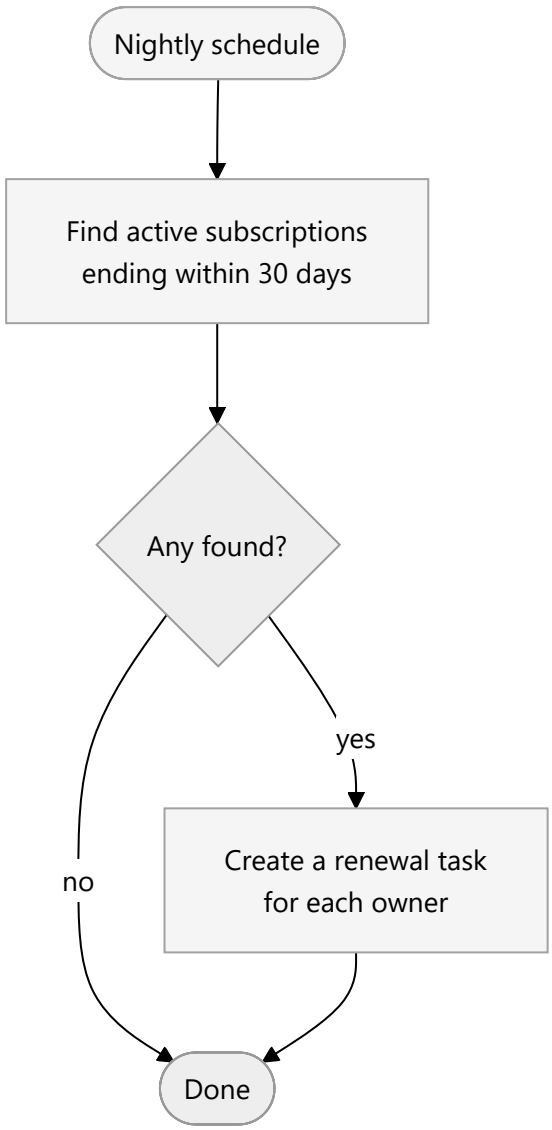


Figure 9. *Renewal reminders*

6.6 Case routing and escalation

Assignment rules route each new case; escalation rules raise cases that miss their target time. These belong to the Service domain.

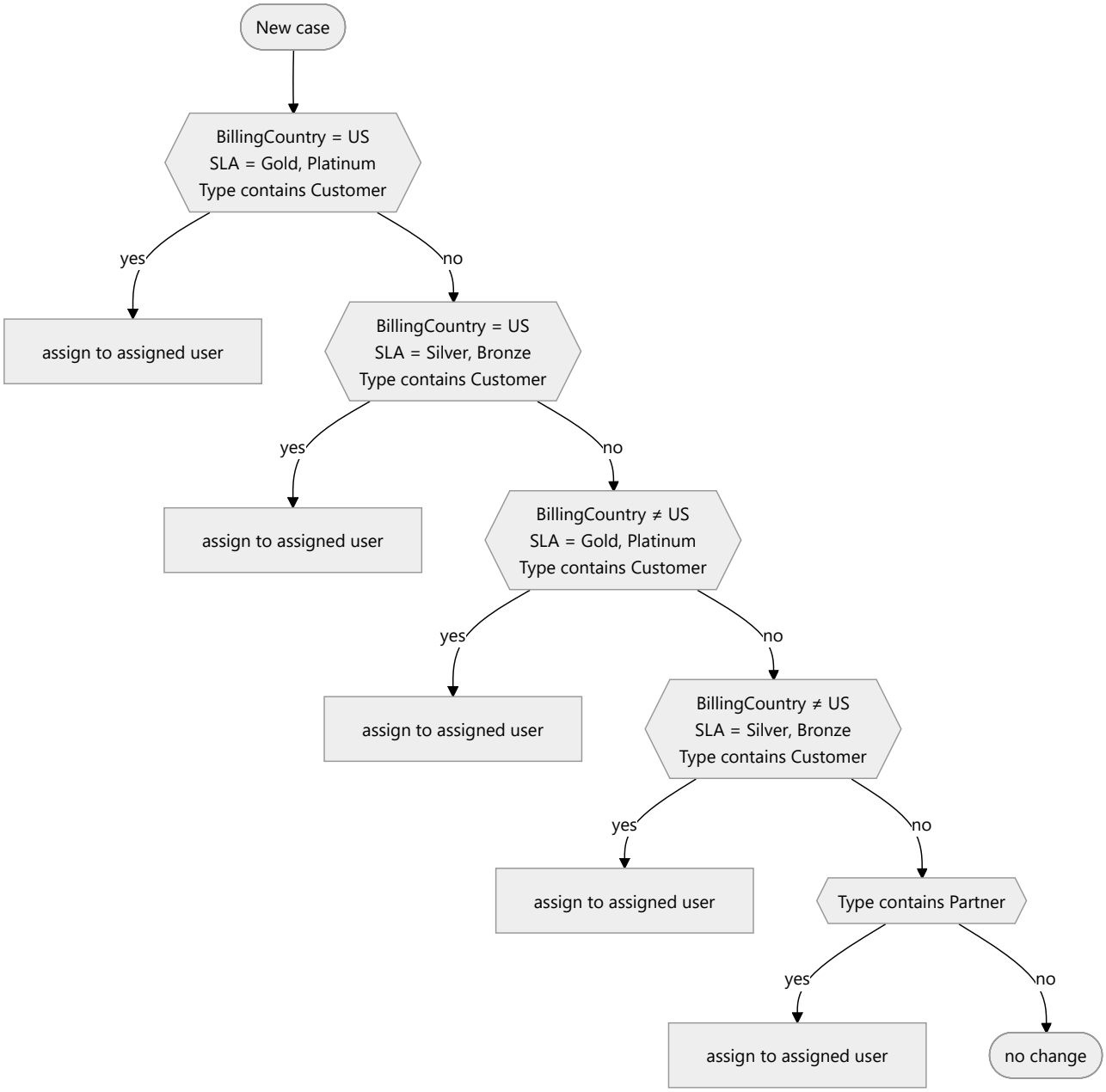


Figure 10. *Case routing and escalation*

Escalation rule	Applies when
1	BillingCountry = US; SLA = Platinum; Priority = High
2	BillingCountry = US; SLA = Platinum; Priority = Medium,Low
3	BillingCountry = US; SLA = Gold; Priority = High
4	BillingCountry = US; SLA = Gold; Priority = Medium,Low
5	BillingCountry = US; SLA = Silver
6	BillingCountry = US; SLA = Bronze
7	BillingCountry ≠ US; SLA = Gold,Platinum
8	BillingCountry ≠ US; SLA = Silver,Bronze

6.7 Working alongside Salesforce

While Sales is still on Salesforce, its changes reach the platform through the change feed; once Subscriptions has moved, its changes go back to Salesforce only where Salesforce features still read them.

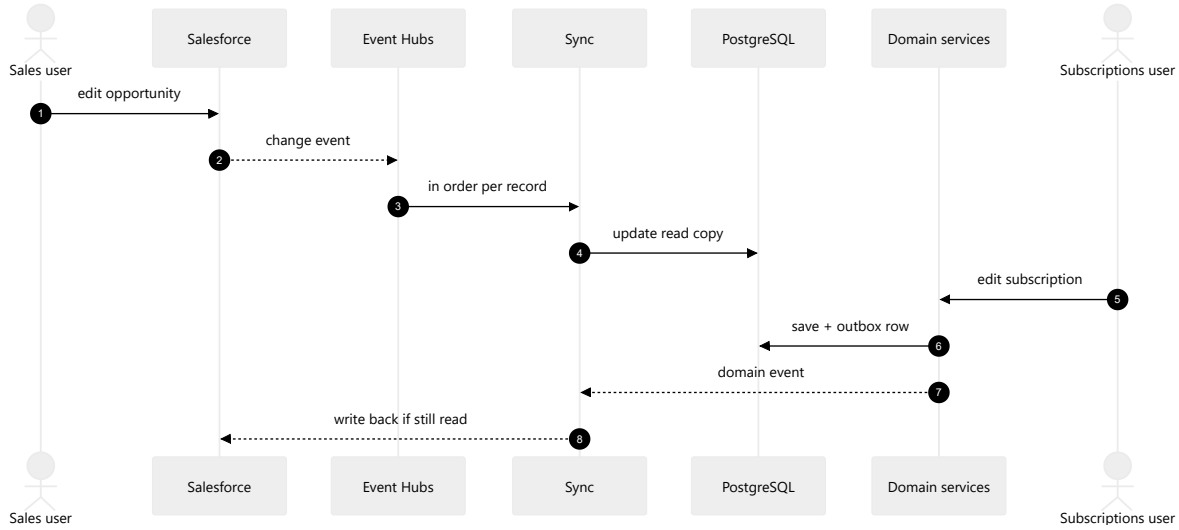


Figure 11. *Working alongside Salesforce*

7. Deployment view

7.1 Target deployment on Azure

One region with three availability zones, a paired region for backups, and private networking in a spoke virtual network connected to the client's landing zone. Every domain runs in the same Container Apps environment.

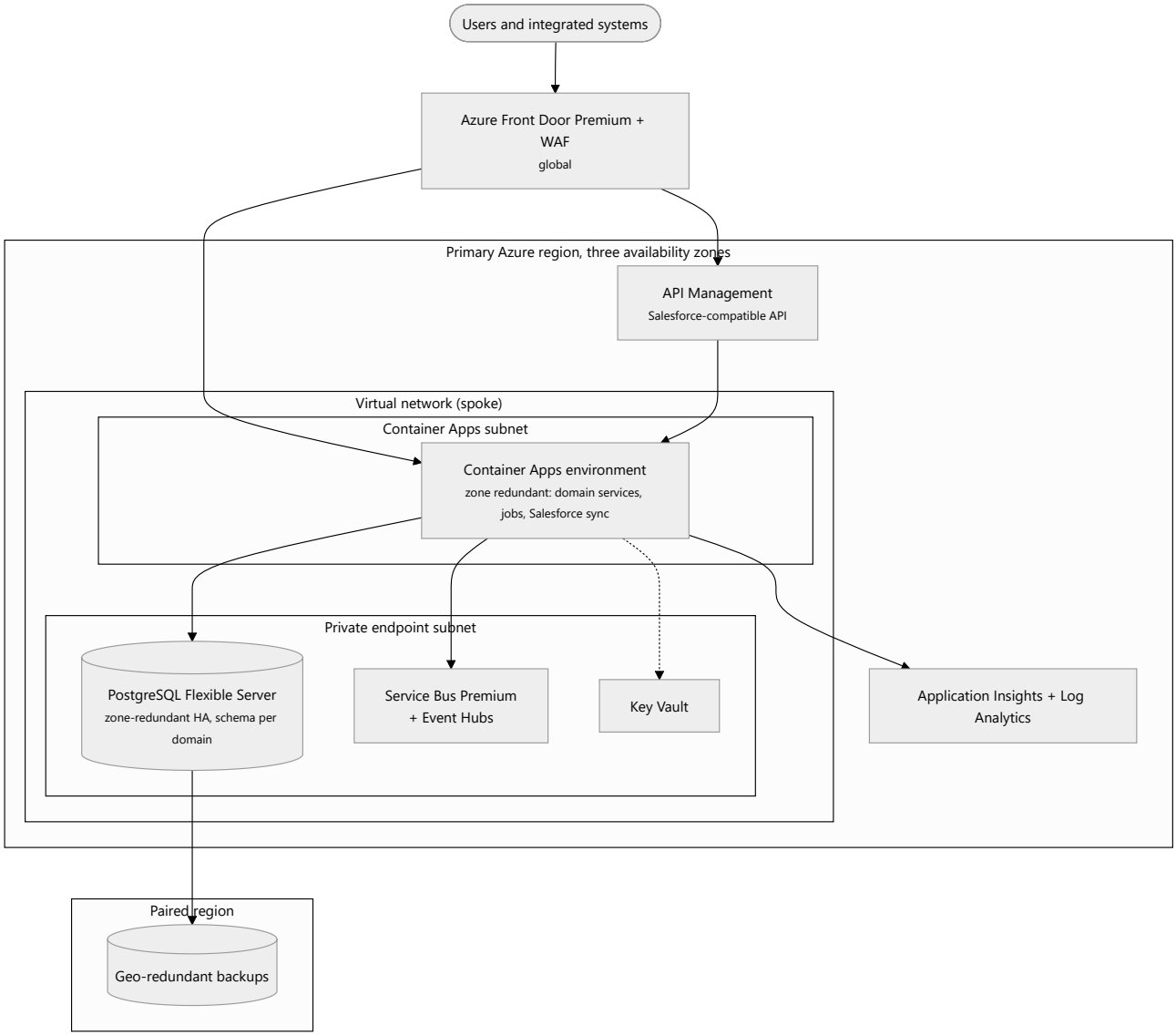


Figure 12. *Target deployment on Azure*

7.2 Services and sizing

Component	Azure service	Production	Proof of concept today
Edge	Front Door Premium + WAF	Managed rule set, bot protection	not deployed
Domain services, jobs, sync	Container Apps	Dedicated workload profile, zone redundant, 2+ replicas, scale rules per app	App Service B1, single instance
Database	Azure Database for PostgreSQL, Flexible Server	General Purpose, zone-redundant HA, geo-redundant backups	Standard_B1ms (Burstable), 7-day backups
Event bus	Service Bus Premium, Event Hubs Standard	Private endpoints, geo-disaster recovery pairing	not deployed
Integration gateway	API Management	Standard v2, VNet integrated	not deployed
Secrets	Key Vault	Private endpoint, RBAC, purge protection	secrets set at deploy time

Component	Azure service	Production	Proof of concept today
Identity	Entra ID, Entra External ID	Single sign-on, conditional access, managed identities	in-app sign-in
Monitoring	Application Insights + Log Analytics	Alerts, dashboards, 90-day retention; Microsoft Sentinel	health check
Infrastructure	Bicep, GitHub Actions	One template per environment: development, test, production	one Bicep template

7.3 Environments and release

Environment	Purpose	Data
Development	Build and unit tests	Synthetic
Test	Parity, security and user acceptance tests	Masked copy of production
Production	Live service	Live

Each release is built once as a container image, deployed as a new revision with no traffic, smoke tested, then given the traffic; moving traffic back is the rollback. Domains release together while they share one deployable.

7.4 Availability and recovery

Event	What happens	Target
Replica failure	Container Apps replaces the replica; others keep serving	no downtime
Zone failure	Replicas in other zones serve; the database fails over to its standby zone	about 2 minutes, no data loss
Region failure	Restore from geo-redundant backup in the paired region	RTO hours, RPO under 1 hour
Bad release	Move traffic back to the previous revision	minutes

8. Cross-cutting concepts

8.1 Data model of the first domain

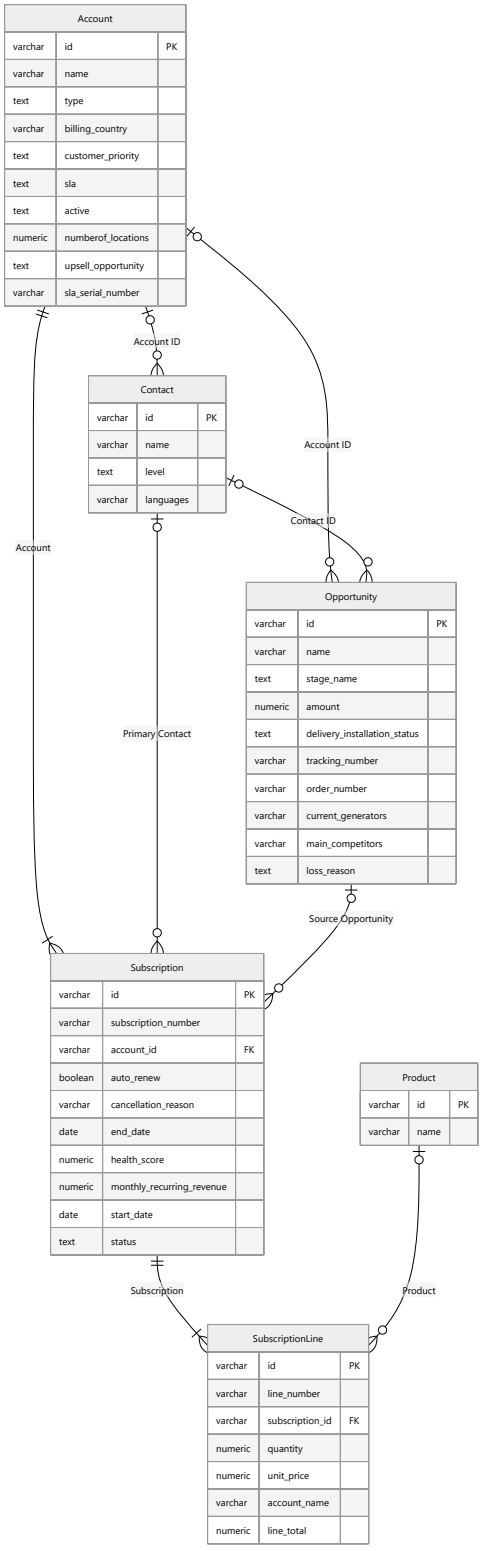


Figure 13. *Data model of the first domain*

8.2 Security across domains

Identity, roles and sharing are one shared model. Every domain schema enforces it with row-level security that calls the same access functions, so a user sees the same records in every module and through every client of the data.

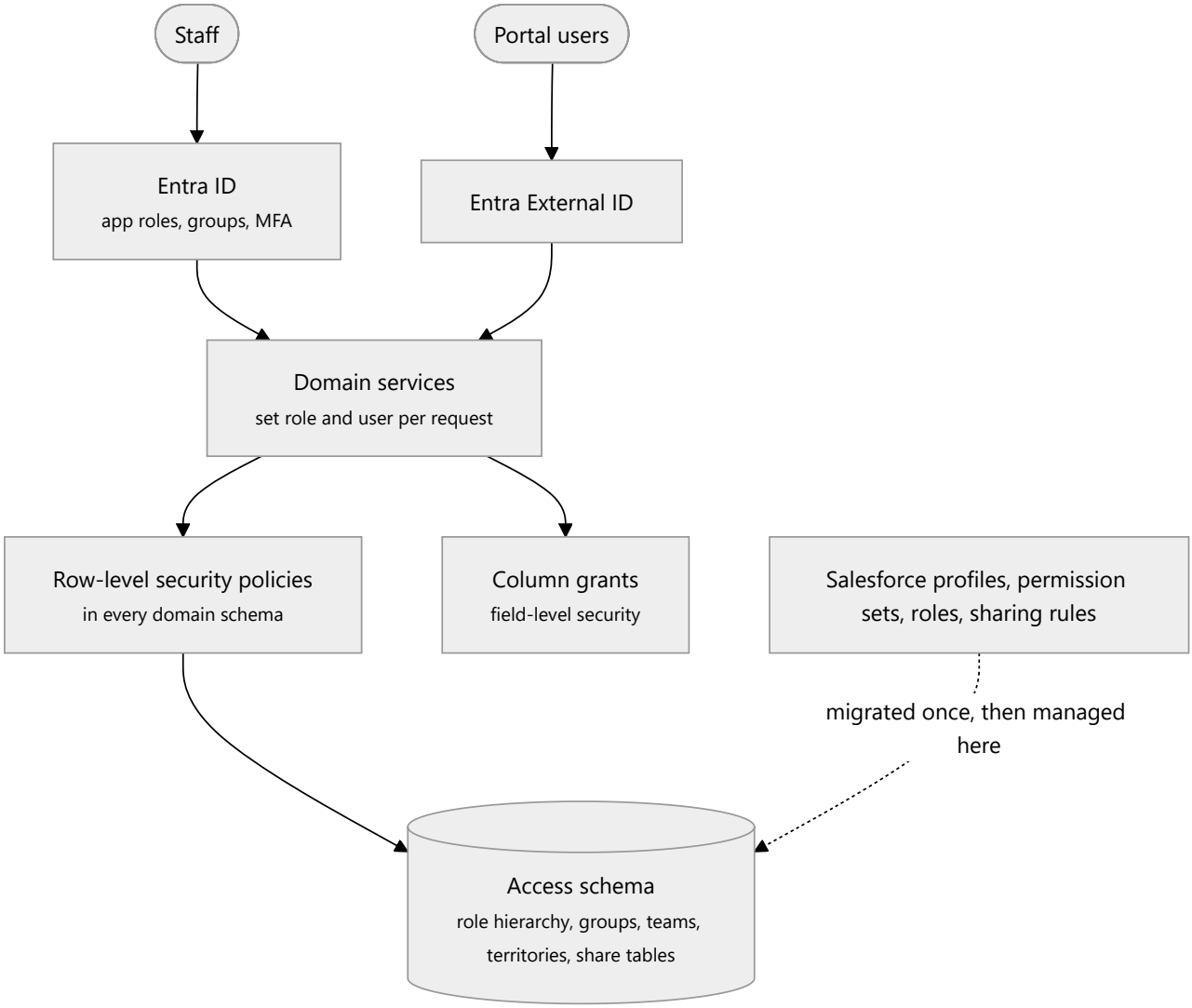


Figure 14. *Security across domains*

Salesforce concept	Target
Users, single sign-on	Entra ID (staff), Entra External ID (customer and partner portals)
Profiles and permission sets	Entra app roles and database roles
Field-level security	Column grants per role
Org-wide defaults, role hierarchy, sharing rules, teams, territories	Access schema (groups, hierarchy closure, share tables) used by row-level security
Shield Platform Encryption	Encryption at rest with customer-managed keys in Key Vault
Field Audit Trail, setup audit	Audit tables and Log Analytics, kept as long as policy requires
Personal data in sandboxes	Microsoft Purview classification; masked copies in test

Roles in the first domain

Each profile or permission set becomes a database role; field-level security becomes column grants; subscriptions inherit access from their account.

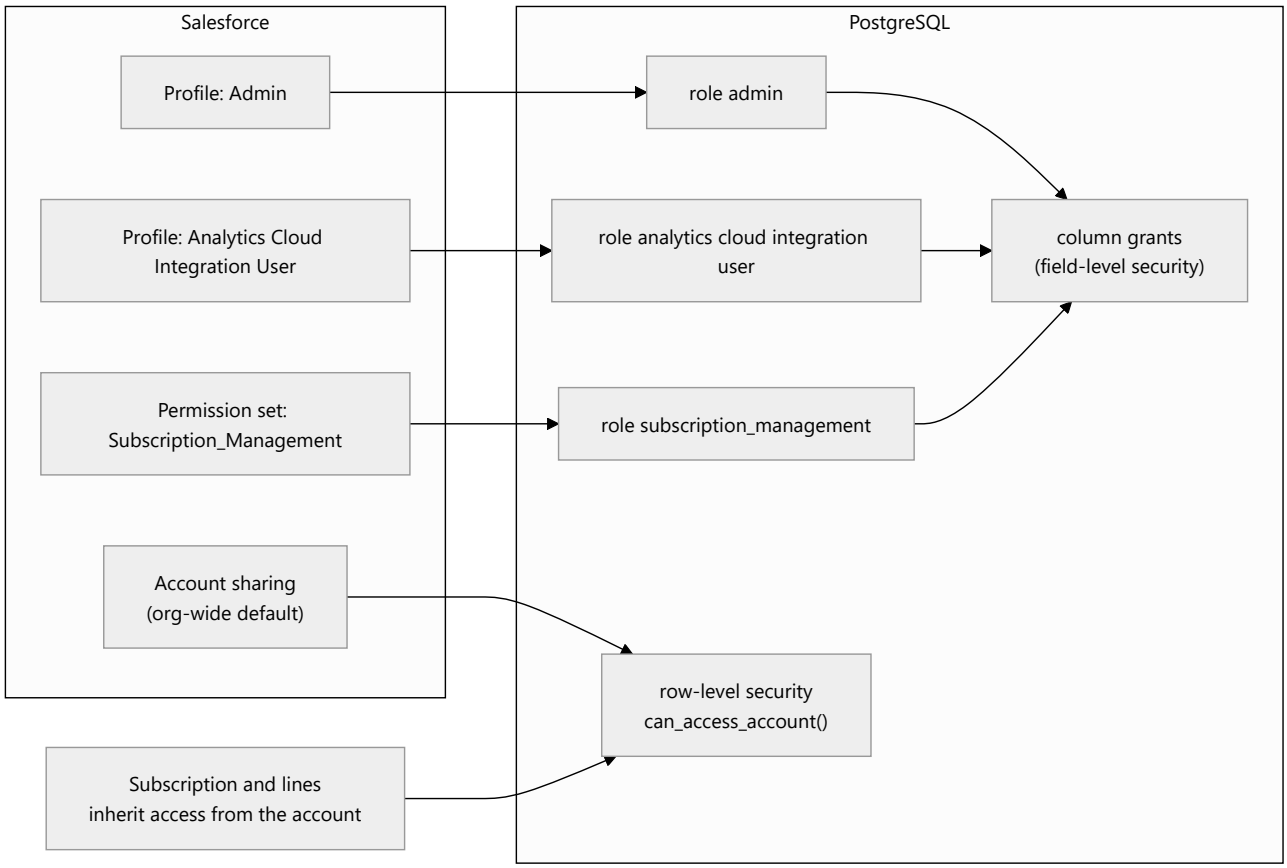


Figure 15. *Roles in the first domain*

8.3 Migration waves and system of record

Domains move one at a time. Core customer data and every in-use domain start as read copies fed by the change feed; a domain becomes master in Azure in its wave; the core moves last, when nothing left in Salesforce writes to it.

Wave	What happens
Wave 0: foundation	Landing zone, platform services, identity; Core customer and every in-use domain as read copies fed by the Salesforce change feed
Wave 1: Subscriptions (pilot)	Subscriptions becomes master in Azure; its users and integrations switch; Salesforce keeps a read-only copy
Wave 2: Sales	Sales becomes master in Azure; its users and integrations switch; Salesforce keeps a read-only copy
Wave 3: Service	Service becomes master in Azure; its users and integrations switch; Salesforce keeps a read-only copy
Wave 4: core and activities	Core customer and activities become master in Azure; Salesforce is read-only
Wave 5: decommission	Final archive taken; licences reduced at renewal

Domain	Wave 1	Wave 2	Wave 3	Wave 4
Core customer	Salesforce	Salesforce	Salesforce	Azure
Sales	Salesforce	Azure	Azure	Azure
Service	Salesforce	Salesforce	Azure	Azure
Subscriptions	Azure	Azure	Azure	Azure
Activities	Salesforce	Salesforce	Salesforce	Azure

Load order in the first domain

Records load parent first, in dependency order, keeping every Salesforce record ID.

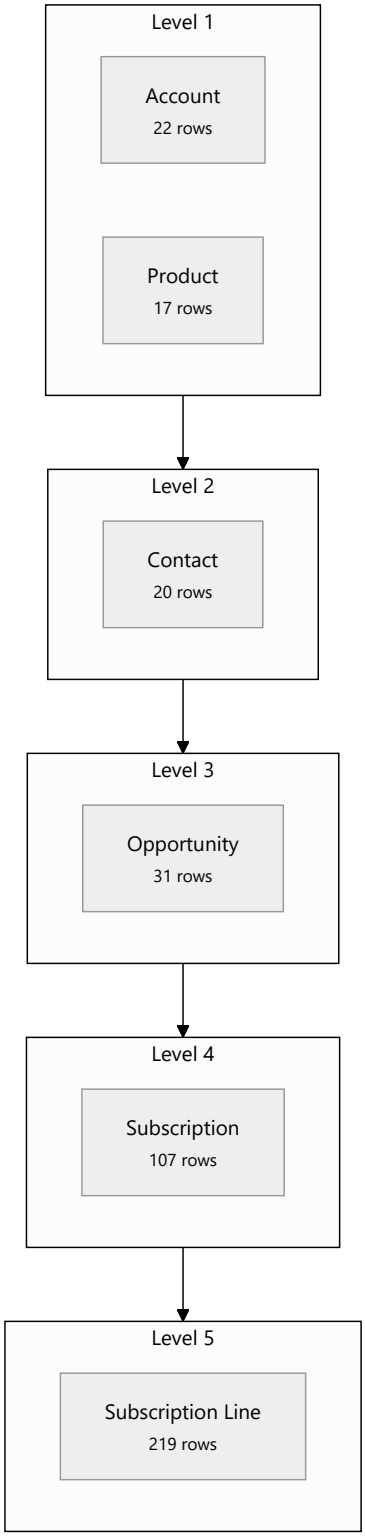


Figure 16. *Load order in the first domain*

8.4 Integration and events

Concern	Standard
External APIs	REST with OpenAPI 3, versioned, through API Management; a Salesforce-compatible facade so callers keep working during the move

Concern	Standard
Events between domains	CloudEvents on Service Bus, written through a transactional outbox; consumers are idempotent
Salesforce change feed	Pub/Sub API (Change Data Capture) into Event Hubs, applied in order per record
Reads across domains	Published read views only; no domain writes into another domain's schema

8.5 Adding the next module

1. Discover and design it as a domain; place it on the domain map and the system-of-record matrix.
2. Generate its schema, rules, access policies and API from the same template; publish its read views and events.
3. Start it as a read copy fed by the change feed; run the parity and security tests.
4. Switch its users and integrations in its wave; Salesforce keeps a read-only copy until decommission.

8.6 Salesforce products outside the core org

Product	Kind	Route
Marketing Cloud Engagement	separate platform	replace with a marketing product, or retain and integrate through the gateway
Commerce Cloud (B2C)	separate platform	retain and integrate, or replace with a commerce platform
Data Cloud, Tableau	separate platforms	replace with Microsoft Fabric and Power BI on the new database
MuleSoft	integration platform	its flows move to API Management, Service Bus and Logic Apps
Managed packages (for example CPQ)	on the core org	settings and data taken; behaviour rebuilt, replaced or retained per package

8.7 Observability

Services and jobs send traces, metrics and logs to Application Insights; Azure resources send diagnostics to Log Analytics; Microsoft Sentinel watches security events. Alerts cover availability, error rate, response time, event backlog, sync lag, database CPU and storage, and failed sign-ins.

8.8 Traceability

The LLD's traceability matrix links each Salesforce element to its target design, database object and test.

9. Architecture decisions

ADR	Decision
ADR-002	Keep the 18-character Salesforce ID as the primary key
ADR-003	Row-local formulas become stored generated columns

ADR	Decision
ADR-004	Cross-object formulas become views
ADR-005	Date-dependent formulas are computed when read
ADR-006	Roll-up summaries are trigger-maintained columns with a recompute job
ADR-007	Row-local validation rules become CHECK constraints
ADR-008	Rules needing the previous row or a parent record become BEFORE-trigger guards
ADR-009	Profiles and permission sets become roles; sharing becomes row-level security
ADR-012	Host on Azure Container Apps and Azure Database for PostgreSQL
ADR-013	Staff sign in with Entra ID, portal users with Entra External ID; services use managed identities
ADR-014	Data services are reachable only on private endpoints; the edge is Front Door with WAF
ADR-015	The platform is split into business domains around a shared core customer domain
ADR-016	Start as a modular monolith: one deployable and one database, one schema per domain
ADR-017	Domains talk through events written with a transactional outbox
ADR-018	Each object has exactly one master at any time; Salesforce and Azure coexist through the change feed
ADR-019	One access model for every domain, enforced by row-level security
ADR-020	Build on open standards so the platform is portable

10. Quality requirements

Quality	Scenario	How it is measured
Parity	Every calculated value and rule outcome matches Salesforce	Automated parity tests, blocking in the release pipeline
Security	A user sees and edits only what Salesforce allows, in every domain	Security tests per role and domain; row-level security and column grants
Modularity	A new module is added without changing existing domains	Contract tests on read views and events; schema permissions between domains
Availability	The service keeps running through the loss of a zone	Zone-redundant services and database; failover drill each quarter
Performance	Screens respond in under 2 seconds at peak load; sync lag under 1 minute	Application Insights and event-backlog alerts; load test before each wave
Operability	Any environment can be rebuilt from source	Bicep templates and the pipeline; no manual changes

11. Risks and technical debt

Item	Status
Cross-domain saves that Salesforce made in one transaction	Listed per wave and covered by parity tests (ADR-017)
Activities refer to records in every domain	Activities move with the core in the last wave
Custom code translated	SubscriptionService.applyDefaults
Automations still to translate	OpportunityTrigger, SubscriptionTrigger, SubscriptionRenewalBatch
Production services not yet deployed	Front Door, API Management, Container Apps, event bus, private networking (section 7.2)

12. Glossary

Business term	Salesforce object	Target table
Account	Account	account
Contact	Contact	contact
Opportunity	Opportunity	opportunity
Product	Product2	product2
Subscription	Subscription__c	subscription
Subscription Line	Subscription_Line__c	subscription_line